

Lesson 24 Inclusion-Exclusion Principle

In this lesson, we learn about the **Inclusion-Exclusion Principle**, which gives a formula for the cardinality of a union of sets.

Let U be the universal set and $A_1, \dots, A_m$ be finite subsets of U . The simplest case of inclusion-exclusion is when we have two sets, then we have the Union Rule.

Union Rule Let A and B be finite sets. Then $|A \cup B| = |A| + |B| - |A \cap B|$.

Proof The LHS counts every element in $A \cup B$ once. On the RHS, $|A| + |B|$ counts the elements which are in both sets twice, so $|A \cap B|$ is subtracted.

“Inclusion-exclusion” means “including $|A|$ and $|B|$, and excluding $|A \cap B|$ ”.

Following are two corollaries from the union rule.

Addition Principle Let A and B be disjoint finite sets. Then $|A \cup B| = |A| + |B|$.

Proof Since A and B are disjoint, $|A \cap B| = |\emptyset| = 0$ and $|A \cup B| = |A| + |B|$.

Subtraction Principle Let A a finite set. Then $|\bar{A}| = |U| - |A|$.

Proof Since A and $\bar{A}$ are disjoint, we have $|A| + |\bar{A}| = |A \cup \bar{A}| = |U|$.

Let's try to find the inclusion-exclusion formula for 3 sets: A, B, C . To find $|A \cup B \cup C|$, we add $|A|, |B|$, and $|C|$, but then we overcount the intersections, so we subtract $|A \cap B|, |A \cap C|$, and $|B \cap C|$. But the elements in $A_1 \cap A_2 \cap A_3$ have been removed: they are added 3 times in $|A| + |B| + |C|$ and subtracted 3 times in $|A \cap B| + |A \cap C| + |B \cap C|$, so we have to add back $|A \cap B \cap C|$,
 $|A \cup B \cup C| = (|A| + |B| + |C|) - (|A \cap B| + |A \cap C| + |B \cap C|) + |A \cap B \cap C|$

Not surprisingly, the **Inclusion-Exclusion Principle** for m sets $A_1, \dots, A_m$ says that:

$$|A_1 \cup \dots \cup A_m| = \sum |A_i| - \sum |A_i \cap A_j| + \sum |A_i \cap A_j \cap A_k| + \dots + (-1)^{m+1} |A_1 \cap \dots \cap A_m|$$

The summations are over all possible intersections.

We give a numerical example before proving the Inclusion-Exclusion Principle.

How many positive integers less than 101 are a multiple of 2, 3, or 5? This is asking for $|A \cup B \cup C|$,

- A is the set of multiples of 2: $A = \{2, 4, 6, \dots, 100\}$
- B be the set of multiples of 3: $B = \{3, 6, 9, \dots, 99\}$
- C be the set of multiples of 5: $C = \{5, 10, 15, \dots, 100\}$

$$\begin{aligned} |A \cup B \cup C| &= |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C| \\ &= \left\lfloor \frac{100}{2} \right\rfloor + \left\lfloor \frac{100}{3} \right\rfloor + \left\lfloor \frac{100}{5} \right\rfloor - \left\lfloor \frac{100}{6} \right\rfloor - \left\lfloor \frac{100}{10} \right\rfloor - \left\lfloor \frac{100}{15} \right\rfloor + \left\lfloor \frac{100}{30} \right\rfloor \\ &= 50 + 33 + 20 - 16 - 10 - 6 + 3 = 74 \end{aligned}$$

Proof of Inclusion-Exclusion Principle

There is an equivalent (complement) version of the inclusion-exclusion example:

Let $A_1, \dots, A_m$ be subsets of the universal set U . Then

$$\begin{aligned} |\overline{A_1} \cap \overline{A_2} \cap \dots \cap \overline{A_m}| &= |\overline{A_1 \cup \dots \cup A_m}| = |U| - |A_1 \cup \dots \cup A_m| \\ &= |U| - \sum |A_i| + \sum |A_i \cap A_j| - \sum |A_i \cap A_j \cap A_k| + \dots \\ &\quad + (-1)^m |A_1 \cap A_2 \cap \dots \cap A_m| \end{aligned}$$

We will prove the complement version, as it is easier to prove. The LHS counts the number of elements which are not in any A_i . We will show that the RHS counts the same thing:

- An element x NOT in any A_i makes a net contribution of 1 to the RHS:
 - x contributes 1 to $|U|$ because x is an element of the universal set (every element is in U).
 - x makes no contribution for the rest of the RHS as x is not in any A_i 's.

So, x makes a net contribution of 1 to the right side:

$$1 - 0 + 0 - \dots = 1$$

- An element y in at least one A_i makes a net contribution of 0 to the RHS:

Suppose y is in exactly n of the m A_i 's, then:

- its contribution to $|U|$ is $1 = \binom{n}{0}$ [It will become clear why we use $\binom{n}{0}$]
- its contribution to $\sum |A_i|$ is $n = \binom{n}{1}$ because n of A_i 's have y
- its contribution to $\sum |A_i \cap A_j|$ is $\binom{n}{2}$: there are $\binom{n}{2}$ pairs of subsets with y
- its contribution to $\sum |A_i \cap A_j \cap A_k|$ is $\binom{n}{3}$: there are $\binom{n}{3}$ ways to choose 3 subsets with y

We continue until the intersection of n subsets. Any intersection of more than n subsets cannot contain y because y is only in n subsets. So, the contribution of y to the RHS is:

$$\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \dots (-1)^n \binom{n}{n} = 0$$

The last equality is the identity proved in #4 of Lesson 12 Exercises.